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Understanding the barriers to the diffusion of rooftop solar: A case study of Delhi (India)

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ABSTRACT

The threat of climate change has necessitated that nations make a switch to greener and more environmentally sustainable fuels. India's plans to promote electricity generation from solar energy is a positive step in this direction. Several states have come up with their own renewable energy plans to support this transition. Delhi's solar policy (2016) intends to contribute to this national plan. It was widely believed that Delhi's favourable geographical conditions for solar, relatively high literacy rate and good public awareness would aid good market penetration of rooftop solar (RTS). However, the expectations of the policy makers have been belied by a meagre adoption rate as RTS has met with a very lukewarm response from the residents of Delhi. The current study holistically explores the obstacles hindering the growth of Delhi's RTS market by integrating perspectives from solar vendors, implementing agencies' officials and users/potential users and suggests some policy measures to address the issues.

1. Introduction

The question of energy in India intersects with several other issues like economic growth, equitable distribution and environmental sustainability. For a developing nation like India, where a substantial proportion of the population lives below or close to the poverty line, reliable energy access is imperative for economic growth. Also, while India's carbon emissions have increased significantly in the recent past, its per capita emissions are on the lower side (World Bank data). India's energy plan must try to address and seek balance with respect to these above issues. India's recent push for renewable energy is an attempt in this direction. Globally, solar is beginning to be considered as a viable renewable energy alternative to the conventional fossil fuels. In 2017, the installed global capacity for solar PV stood at 403 GW (Masson and Kaizuka, 2019) and the solar industry is poised to become one of the fastest growing industries in the world because of the price drop in the solar modules that would help its penetration to the new markets (Bazilian et al., 2013). While India's current energy paradigm is still coal dominated, with coal accounting for 54.2% of the total installed capacity (Power Sector at a Glance: All India), India has launched several ambitious renewable energy programs in the last decade with a special emphasis on promoting solar energy.

Reflecting their unique advantages and challenges, different states in India have come out with their own renewable energy policies to contribute to the national renewable energy plans. Delhi also formulated its solar policy in 2016 according to which Delhi aims to generate close to 1 GW by the year 2020 (see Table 1). However, unlike states like Gujarat, Rajasthan and Karnataka, that have been relatively successful in achieving solar energy penetration (Harish et al., 2013; Pandey et al., 2012), Delhi is nowhere close to achieving the targets set in the policy. The study attempts to fill a literature gap as apart from some brief articles in the press, to the best of the author's knowledge, no holistic empirical study of solar penetration has been done in Delhi.

The next section gives a brief sketch of India's solar programs. The following section, i.e. section 3 elucidates the relevance and the state of RTS in Delhi. The method employed in the study is given in section 4. Section 5 gives an account of RTS diffusion barriers in Delhi and the conclusion and some policy suggestions are provided in section 6.

2. Solar power programs in India: A brief sketch

India receives an average solar radiation intensity of about 200 MW per kilometer square (MW/km^2) with 250–300 sunny days in a year which translates into 5000 trillion kilowatt hour (KWh) per year (Sharma et al., 2012). Recognizing the potential of solar in India's energy future, the government of India launched Jawaharlal Nehru National Solar Mission (JNNSM) in 2010, which was one of the first ambitious programs to promote solar energy at that time (Sharma et al.,

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Table 1
Yearly targets set in Delhi solar policy (2016) (Source: Delhi solar policy, 2016).

Fiscal year	New solar energy (MW)	Cumulative solar energy (MW)	Annual growth (%)	Percentage of peak grid load	Percentage of total electricity consumption
2016	30	35	700	1	0.15
2017	84	119	240	2	0.56
2018	193	312	162	5	1.43
2019	294	606	94	9	2.66
2020	385	991	63	14	4.16
2021	285	1275	29	17	5.10
2022	228	1503	18	19	5.73
2023	187	1690	12	20	6.14
2024	161	1850	10	21	6.40
2025	145	1995	8	21	6.57

2015). The target for solar generation according to this program was set at 20 GW by 2021–22 (PIB, 2015). However, according to the recently revised targets, the government of India is committed to generate 175 GW of renewable energy by 2021–22 (PTI, 2017), of which 100 GW is aimed to be generated through solar (PIB, 2015). RTS has been given specific attention in India's solar policy. It is a part of JNNSM as 'Rooftop PV and Small Scale Generation Program' (RPSSGP). Out of the 100 GW of power to be generated through solar till 2021–22, the government is aiming to generate 40 GW from RTS projects. Studies like Sundaray et al. (2014) claim urban RTS potential in India to be at 124 GW, of which 38% can be contributed by the residential sector. However, studies so far show that while the ground mounted solar installations have picked up some pace, the RTS segment is lagging behind (Engelmeier et al., 2013). For a review of the challenges and trends in the rooftop solar segment in India see Goel (2016).

3. RTS in Delhi

3.1. Rationale for RTS

The federal structure provided in the constitution of India encourages the states to have a proactive approach towards energy sector policies for several reasons. First, electricity, being a concurrent subject in the constitution of India, requires an active cooperation between the centre and the state governments. Second, The Electricity Act (2003) mandates governments at different levels to promote the generation of renewable energy. Third, states must develop energy plans that reflect their natural conditions and availability of resources. For instance, Delhi has a favourable geography for harnessing solar power which has not been fully utilized so far. Finally, it is a well-established fact that urban centres like Delhi have a high energy demand (Abosedra and Baghestani, 1989) and globally, are responsible for consumption of approximately one-third of the total fossil fuels and about 80% of all the carbon emissions (Vince, 2013). Thus, urban centres like Delhi, should be one of the foci of sustainability experiments.

Delhi is a landlocked region with scarce availability of contiguous land parcels and has meagre wind and hydro energy resources (Delhi solar policy, 2016). It has a lot of ground to cover in the sustainability area as it was ranked among the bottom five in Sustainable Cities Index (2015) which used energy efficiency and environmental pollution as components of its multi criteria assessment (Sattiraju, 2016). However, it has favourable conditions for harnessing solar energy. It receives an average solar irradiation of about 5 KWh/m^2/day. Such an amount is considered adequate although some African nations get about 7–8 KWh/m^2/day and in India, certain states like Rajasthan get 6–7 KWh/m^2/day. Almost 300 sunny days that Delhi gets in a year, combined with a rooftop space of about 31 km^2 gives it an annual solar potential of approximately 3500 million KWh (Delhi solar policy, 2016). Further, owing to the high degree of urbanization, RTS is considered to be the most appropriate and viable of all renewable energy options in Delhi (BRPL's Solar City Initiative). According to some calculations, just by using 1.6% of its available roof space, Delhi can generate about 2 GW of solar power (Engelmeier et al., 2013). This largely untapped market of RTS presents a lucrative market opportunity for the business sector and a significant occasion for Delhi to contribute to the national transition to sustainable energy. Moreover, since energy produced by RTS systems is consumed mostly at, or near, the point of generation, it would help in minimizing transmission and distribution losses (Delhi solar policy, 2016).

3.2. Current state of RTS in Delhi

To harness the RTS potential and implement its vision, the union government felt that distribution companies (DISCOMS) would be the appropriate implementing agencies since they have the direct access to the consumers through an organised billing system (Lok Sabha unstarred question No. 4128). Ministry of New and Renewable Energy (MNRE) of the union government has therefore accorded a sanction of 10 MW capacity each to three DISCOMS operating in Delhi - BSES Rajdhani Power Limited (BRPL), BSES Yamuna Power Limited (BYPL) and Tata Power Delhi Distribution Limited (TPDDL) - for RTS programs implementation (Krishan, 2020). Thus, BRPL in collaboration with The Energy and Resources Institute (TERI) and Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ India) under its Indo-German Solar Partnership project and with the support from the governments of Delhi and India, launched Solar City initiative in Delhi. The initiative aims at optimally utilizing the untapped potential of RTS in Delhi by installing solar panels on roof spaces in Delhi. It would also help the DISCOM in meeting its Renewable Purchase Obligation (RPO) (Rajeshwari, 2018) as Delhi solar policy (2016) mandates that DISCOMS must give priority and preference to sourcing at least 75% of their RPO targets from within the state of Delhi. Electricity generated through the DISCOMS' RTS programs would count towards their RPO obligations. Also, RTS can help the DISCOMS in reducing peak demands as Delhi's day time peak demand curve broadly matches the generation curve of solar system (Delhi solar policy, 2016). Thus, it may help the DISCOMS in minimizing overloading issues in the congested areas during the peak summer months (BSES press release, 2018).

A considerable proportion of RTS potential in Delhi is present on the rooftops of approximately 1990 registered Co-operative Group Housing Societies (CGHS) in Delhi (Mukhya Mantri Solar Power Yojana, 2018). Earlier, Delhi government's efforts to promote RTS on individual homes wasn't successful due to paucity of space. Since housing societies' roofs have ample space for big installations, therefore, it is believed that CGHS would be a good segment for the RTS program in Delhi and based on the response, the program could be expanded in a systematic manner to other residential segments as well (BSES press release, 2018; R-16, government/DISCOM official; R-26 government/DISCOM official). Since Dwarka region in west Delhi has a good density of CGHS, therefore on January 7, 2018, BRPL launched Solarize Dwarka initiative which is the first program of their Solar City initiative. In Solarize Dwarka, BRPL is focusing to install solar panels on CGHS' roofs, in-line with the DERC Net Metering Regulations of 2014. The initiative is considered unique as it seeks to provide installations at a single point for an entire society (PTI, 2018). The DISCOM would be responsible for creating a platform for bringing together consumers and RTS vendors and would also design programs for educating consumers of the benefits of solar energy while ensuring strict quality compliance of the systems being set up (BRPL's Solar City Initiative).

Currently, there are two models of RTS functional in Delhi – CAPEX and RESCO. In the CAPEX model, the consumer has to finance the cost of system installation and consequently has full ownership rights over the system. The consumer can avail bank loan, however, as it will be pointed out later, it is difficult to get. The vendor provides free maintenance for the first 5 years only and during this period the vendor is required to address any complaints within 48 hours of receiving them. Thereafter, the onus for proper functioning of the system rests with the consumer. The maintenance contract can be renewed for a fee between the consumer and the vendor. In the RESCO model, the installing company bears the cost of the installation in exchange for a bank guarantee, equivalent to three years of generation cost, from the consumer as collateral. The company sells the electricity to the consumer at a predefined rate which is less than the grid price. Operation and maintenance is the company's responsibility for 25 years (BRPL booklet – vendor and contract). The cost structure of the two models is given in the Table 2.

The energy generated through RTS is used for the common areas of the society like lift, pumps and common space lighting and excess electricity is fed into the grid and the electricity bill is adjusted under net metering (BRPL's Solar City Initiative). According to DISCOMS officials, a 1 KW RTS PV system can generate 4–5 KWh of electricity per day which can result in an average monthly savings of up to Rs. 750 for up to 25 years (PTI, 2018; BRPL's Solar City Initiative). At the time of the study i.e. in the financial year 2018–19, the union government through MNRE was giving a capital subsidy of 30% of the RTS installation cost (PIB, 2017). In addition to this, Delhi government was giving a Generation Based Incentive (GBI) of Rs 2.00 per unit on solar generation for a three year period starting from the financial year 2016–17 (Mukhya Mantri Solar Power Yojana, 2018). Since then, the union government has increased the subsidy amount from 30% to 40% (MNRE demand for grants 2020-21).

The government of Delhi describes its solar targets as 'aggressive yet achievable'. It was expected that better understanding of the technology and environmental awareness in cities like Delhi would facilitate the diffusion of solar (Chaudhary, 2016). It was also believed that favourable economics of solar vis a vis conventional power and emergence of innovative business models would make solar a lucrative option for consumers overtime (Engelmeier et al., 2013). However, it is observed that the state government has failed to achieve its targets against the time period stipulated in the policy document (see Table 1). In the two years following the policy launch i.e. 2017 and 2018, only 300 RTS installations were done (Nandi, 2018) and only 5 MW of residential rooftop solar could be added (Goswami, 2018). It is clear that the hopes entertained by the government with respect to solar did not actualize as officials have admitted that the response of domestic/residential consumers to solar has been 'poor' (Mukhya Mantri Solar Power Yojana, 2018).

4. Method and respondents

The study involved 40 semi structured interviews comprising 20 government empanelled solar vendors and 17 housing society members involved in management and decision making and 3 government and DISCOM officials. Talking to different stakeholders enabled the author to view the same issue from multiple perspectives, provided a holistic understanding and offered an opportunity for triangulation of perspectives against each other. For instance, the issue of cost came out as a significant barrier in the study, however, it had multiple aspects that were highlighted by different stakeholders. This multi dimensionality would have been lost if the study would have relied on stakeholders from a single constituency only. Dwarka region of Delhi was chosen for interviewing the consumers/potential consumers as BRPL launched its RTS program Solarize Dwarka here. The program involved convincing housing societies through information dissemination to install solar panels on their roofs. Consequently, the awareness regarding RTS was relatively high in this region and it also provided a good density of respondents for the study. The names of first few respondents were obtained from internet and news articles search. Respondents were explained the purpose and the objectives of the study. Respondents who agreed to be a part of the study were interviewed and were also asked to suggest names of other respondents who could be relevant for the study. Thus the study relied on snowball sampling.

Interviews were semi structured but focused on certain themes. For solar vendors and government/DISCOM officials, the interview focused on the themes of dynamics and structure of solar market in Delhi, working of government institutions, organizational and financial aspects of their enterprise and Delhi government's solar policy (2016). For housing society members, the interview focused on their motivation for considering or not considering RTS, information and awareness regarding RTS, finance provisions, interaction with government officials and vendors and effectiveness of the government's steps in promoting solar. Interviews were recorded and transcribed except in two cases where the respondents did not allow recording. In these two cases, notes were made manually. Respondents have been given alpha numeric codes to protect their identity.

The method of thematic analysis was used in analysing interview transcripts. Thematic analysis basically involves examining data for prevalence of themes and is a common method for analysing interview or newspaper data (Joffe, 2012). Thematic analysis identifies and enables the researcher to analyse recurring patterns of expressions in her data. But it often goes further and helps in interpreting different facets of the research topic in a general way (Braun and Clarke, 2006). Joffe (2011) argues that thematic analysis is a tool through which both manifest and underlying meaning of the data can be systematically analysed by capturing the nuances people bring to their expressions. Themes were inductively identified from the data. This means that themes were not driven by any prior theoretical frame thought of by the author. The data was transcribed and read multiple times to identify the themes. This means that themes bear a strong association with the data themselves rather than the researcher's theoretical perspective (Patton, 1990). However, this doesn't mean that inductive thematic analysis is done in an 'epistemological vacuum'. Researcher's theoretical and epistemological values are always part of his/her research (Braun and Clarke, 2006). However, there was no conscious attempt to fit the data into any pre given theoretical frame. The transcripts were coded manually and extracts were analysed to look for salient expressions or the aspects that were focused upon by the respondents. The themes extracted from the responses helped in identifying the issues posing a challenge to the effective dissemination of rooftop solar in Delhi.

5. RTS diffusion barriers in Delhi

Diffusion and adoption of a new technology like RTS requires an effective institutional support system – comprising of both formal and informal institutions – to be in place. As institutions and technology co-evolve and adjust to each other in the long term, the initial stages of technology diffusion are marked by uncertainties (Painuly, 2001) which need to be addressed through adequate interventions. The study showed several barriers hindering RTS diffusion in Delhi which are discussed below. For a concise tabular formulation of the different dimensions of the barriers and policy suggestions, see Table 4.

5.1. Cost

Prohibitive effects of perceived high prices on diffusion of solar are well recognized in the global literature (Zhang et al., 2012) and

Table 2
Cost structure of CAPEX and RESCO models (Source: http://solarbses.com/download.aspx).

CAPEX		RESCO		
Size (KWp)	Net cost per KWp in INR (after subsidy)	Size (KWp)	1st year tariff per KWh (INR)	Levelized tarrif with escalation at 3% per annum (INR)
1–10	29,400	25–50	3.13	3.93
10–100	34, 930	50–100	3.15	3.95
>100	32, 900	>100	2.85	3.58

corroborated from other Indian states' analysis of RTS programs (Kappagantu et al., 2015). Thus, it was no surprise that the issue of cost came out to be as the most significant barrier hindering the diffusion of RTS in Delhi. The issue has multiple facets and needs to be understood from the perspective of the consumers as well as the vendors. From the consumer's perspective, the installation cost under the CAPEX model was perceived to be high by many societies even after factoring in government subsidies. Vendors mentioned that many installations were a result of personal contacts and references (R-7, vendor) and very few discussions about solar go beyond preliminary enquiry stage because of the cost (R-6, vendor). Depending on the size, an installation under CAPEX model could cost anywhere between Rs. 2 and 6 million, which most of the societies found to be in excess of their financial capacity. For instance, R-24 (society member) mentioned that their society could only gather Rs. 1.7 million out of Rs. 5.6 million needed while R-36 (society member) recalled that many people in the society's general body meeting objected to solar on hearing the installation cost of Rs. 2 million. Similar issues were faced by other societies who found installation cost 'out of reach' and a 'return on investment of Rs. 400 per capita per month' to be not tempting enough (R-29, society member). It must be noted that cost wasn't a problem for everyone as one society in the entire sample commented on the installation cost as 'no problem at all' (R-17, society member). However, for most of the societies, the installation cost was a big obstacle which is adequately expressed in the following excerpt from R-33's (society member) transcript:

> ' …. it is simple …. our main problem is that the cost of solar installation is still very high'

The RESCO model which requires no upfront cost by the consumer is considered to be a good solution by the government officials to the CAPEX model installation cost issue. However, the model was perceived as risky by vendors while many societies didn't find it appealing enough in terms of the returns on investment. Compared to the CAPEX model, very few vendors were willing to install under the RESCO model because it is believed that it puts a lot of risk on the vendors (R-16, DISCOM/government official). For societies too, it was mostly a 'second choice' (R-24, society member) as they found it to be 'less profitable than the CAPEX model' (R-17, society member). Another issue pertaining to the RESCO model was that it required a certain threshold of consumption to make it economically feasible for the vendors. For instance, for R-36 (society member) the CAPEX model presented 'money issues' and for the RESCO model, they did not have 'enough consumption'. Thus, he was not able to find any vendors. On the other hand, R-23 (society member) had RTS installed under the RESCO model but their first preference was also the CAPEX model, but they abandoned it due to insufficient funds. He mentioned that under the RESCO model their society was generating about 10,000 units monthly since last 8 months but they were exporting very little to the grid, only about 500 units monthly. Their monthly savings came out to be approximately Rs. 100 per capita.

The issue of cost was a grave concern for vendors too. Many vendors felt that the bidding cost of some vendors was unsustainable and was distorting the market (see Table 3). For, example R-9 (vendor) found it 'disturbing for small solar businesses' that the lowest bid price per unit was about Rs. 36. He calculated the price as:

> ' … …look here …. .per unit price of solar panel comes to be around [Rs.] 26–28 … … for wiring, earthing, installation, it is around [Rs.] 12–14 …. .for small inverter …. [Rs.] 9–10. In my opinion, it [per unit price] should be around [Rs.] 50'.

Other vendors like R-14 too 'couldn't understand the pricing' of the lowest bidders, while R-12 (vendor) suspected whether some vendors have been able to reduce prices 'through unfair means'. R-11 (vendor) believed that the scenario was 'disturbing' and was contemplating on venturing into some other business since he thought that there was 'no other option'. Moreover, since 30% capital subsidy by MNRE is decided on the basis of the lowest bidding cost, the 'unfairly' low cost also led to lower capital subsidy for solar users. In addition to this, vendors mentioned that the low bidding price was causing mistrust between vendors and potential consumers. R-13 (vendor) shared his experience in this matter. He recalled that several interested clients 'would compare my price with the lowest price and would feel that I am charging them unfairly'. The author talked to the lowest bidder and found that since they claimed to be an NGO, they could 'claim certain tax exemptions like on custom duties etc.' and by using CSR money, they were able to 'partially offset the cost' (R-5, vendor). This is how they were able to bid a low price. While there may be nothing 'unfair' in this mechanism, the discontent among vendors over pricing and the decision of vendors to exit the 'distorted' solar market is an issue that needs to be examined.

The third aspect of the cost issue that hinders RTS diffusion in Delhi is the relative cost of solar vis a vis conventional grid electricity. The attractiveness of RTS as an electricity generating option depends on its cost competitiveness with respect to the grid electricity. If the grid electricity prices come down, then solar would start losing its economic advantage. This is a significant factor in the RESCO model which involves signing of a long term contract between vendors and consumers. If the grid electricity tariff is revised downwards after a consumer commits to a RESCO model, then s/he 'will feel cheated' (R-26, DISCOM/government official). Vendors also pointed out this issue as 'a challenge of the RESCO model' as if the grid prices undergo downward revision then their consumers 'may start questioning the installed system and may default on the payments' (R-25, vendor). Providing cheap or free grid electricity has been a political governance issue in Delhi and thus governments often feel obligated to fulfil this promise (PTI, 2019). However, this can have an adverse effect on the RTS market in Delhi. Delhi government's own official document recognized this adverse effect as it admitted that new lower grid tariffs announced by DERC in November 2017 reduced the relative advantage of solar power in Delhi (Mukhya Mantri Solar Power Yojana, 2018).

5.2. Finance

Some form of financial support is imperative for renewable energy diffusion in developing nations and lack of it can obstruct the budding market in the initial stages itself (Chaianong and Pharin, 2015). In the study, it was found that the cost issue was aggravated by the lack of finance for the customers interested in RTS. This problem was recognized by all the stakeholders – vendors, societies and DISCOM officials. Housing societies often don't have any collective mortgage or collateral to offer to the banks to avail solar loan. Societies in such a situation often meet a dead end. R-24 (society member) found himself in a similar situation where banks 'flatly refused' them any loan as they did not have any collateral. Even the offer of 'repaying the bank from the savings they would make out of the RTS system didn't change the bank's mind'. R-5 (vendor) recounted that one of his customers, 'an ex-army man, tried so hard to get a loan from a bank but finally gave up after two months'. DISCOM officials understood the 'seriousness of the problem' and assured that 'it was on their mind' but also admitted that as of now 'no bank will give a solar loan to a housing society because there is no collateral' (R-26, DISCOM/government official). Interestingly, R-31 (society member), a banking official himself, was able to avail a bank

Table 3
Lowest bids in different category sizes under the CAPEX model. Source: BRPL's Solar City Initiative (http://solarbses.com/download.aspx).

Size (in KWp)	Lowest bid (per KWp in INR)
1–5	35875
5–10	43323
10–500	38500

Table 4
Summary table for barriers and policy recommendations.

Barrier	Explanation and aspects	Cause(s)	Effect(s)	Suggestion(s)
Cost	Installation cost perceived high by consumers Bidding cost perceived low by vendors Low prices for grid electricity hurt competitiveness of solar	Inadequate funds with societies Some vendors bidding at 'unsustainable' price Political/governance obligation causing government to reduce grid electricity prices	Demotivates consumers Discourages vendors and breeds mistrust between vendors and customers Makes solar relatively less attractive	Try to lower prices through technological and financial innovations Resolve vendor's grievance about bidding price through interaction Adopt rational and not political approach to pricing
Finance	Lack of financing options for RTS	Absence of collateral with societies to avail loan Cumbersome loan process with no specific products for solar	Even interested consumers give up on solar	Streamline loan process Introduce solar specific financing products
Information	Inadequate information among public	Vendor's perception of government agencies not promoting solar enough Society members' perception of vendors 'hiding' information.	Less awareness about RTS′ benefits Delays decision making Trust issues between consumers and vendors	Launch an expansive awareness campaign Facilitate consumers-vendors platform to exchange information and build trust
Subsidy disbursement process	Vendors perceive the process as tedious	Cumbersome documentation Indirect subsidy transfer to consumers through vendors	Demotivates vendors Shift from residential to unsubsidized commercial segment	Online documentation Direct subsidy transfer to consumers
Trust	Society members' suspicion regarding vendors	Members suspect vendors of not sharing required information	Credibility loss for vendors and RTS in general	Facilitate consumers-vendors platform to exchange information and build trust
Other barriers	Slow decision making in housing societies Uncertainty about the future of societies' RTS decisions Revenue loss to DISCOMS	Members face difficulty in gathering information and convincing other members Vendors' fear that new elected members may not abide by old member's decisions about solar High paying customers switch to solar and tariff structure issues	Discourages consumers Discourages vendors Discourages utilities from promoting solar effectively	Provide help to societies to quicken decision making process Facilitate information and experience sharing among societies Involve a trusted entity like DISCOMS to facilitate agreements and penalize in case of violation Rationalize tariff structure to minimize DISCOMS′ loss

loan for solar on behalf of his society. However, he asked the author during the interaction to not reveal any details about the loan.[1] The issue of lack of finance has been given some attention in the press and is a problem for societies in other areas of Delhi as well, where several housing societies gave up on installing solar when they failed to arrange loans (Tiwari, 2019).

5.3. Information

Dissemination of any new technology is predicated on its social acceptance which in turn depends on the degree of public awareness of its benefits (Wüstenhagen et al., 2007). Thus, in absence of adequate information, market penetration of a novel technological intervention like solar would naturally be limited (Painuly, 2001). This study found that the lack of information was a big issue for both vendors and society members albeit in a different manner. For vendors the information barrier was about the lack of government efforts in disseminating general information about solar among people. For society members, the issue manifested in a failure to get adequate technical and useable information from the vendors which could aid in the decision making. Newspaper reports too mention the lack of awareness of commercial and environmental benefits of solar among general public in Delhi. For instance, Goswami (2018) found that even some of those who had installed RTS systems were not aware of its environmental benefits and were not availing its full commercial gains. While vendors maintained that public had general awareness, they nevertheless felt that there was not enough information for them to install solar. In absence of knowledge about its benefits they felt that 'people will not opt for an expensive option like solar' (R-11, vendor). Some of the vendors blamed the government for reneging on its responsibility of promoting solar. R-12 (vendor) criticized the government on not making good on its promise of actively utilizing radio, television, newspaper and other communication platforms to promote solar. According to him, 'people have no idea about its [rooftop solar] benefits or requirements'. In a similar vein, R-9 (vendor) stated:

> 'IPGCL [Indraprastha Power Generation Company Ltd.] promised mass marketing …. but nothing happened'

The conspicuous frustration among the vendors regarding lack of awareness was reflected in R-7's (vendor) angry rhetorical question:

> 'Tell me, what has Government of Delhi and IPGCL done for promoting solar rooftop?'

Another vendor who installed an RTS in a society under the RESCO model complained that DISCOMS were not doing enough to actively promote solar. He told that the DISCOM was not able to facilitate consumer-vendor interaction in his case and he got his clients through his own channels. He added that the DISCOM wasn't even aware that he was in contact with societies till he put his application for net metering (R-25, vendor).

From the residents' perspective information gathering was a significant obstacle in engaging with the government's RTS program. Most of their ire was directed at the vendors for not sharing the information they needed and less towards the government. R-32 (society member) felt that his repeated attempts to elicit the required information from the vendors met with a 'lukewarm response from them'. R-27 (society member) similarly complained of not getting a 'proper response' from the vendors. In a similar vein, R-36 (society member) felt 'very confused' as 'there was no proper dissemination of information'. He complained that he was trying to get some information about the cost and the subsidy structure of the system but was not being given the required information. The absence of useable information for decision making not only stalled but also sometimes led society members to completely give up on the project for good. In addition to this, it also led to breeding of mistrust between the vendors and the society members. This is taken up in section 5.5.

[1] The respondent called the author two days after the interaction to remind that his name must not be disclosed in the context of the solar loan.

5.4. Capital subsidy disbursement process

During the year of the study, union government was offering 30% capital subsidy on RTS installation. Most of the respondents felt that the subsidy was required for market growth, especially in the initial stages, however, they were dissatisfied with the cumbersome and tedious process of availing it. The main issue that vendors flagged regarding the process was its indirect nature. Though the subsidy was meant for the customers, instead of giving it to them directly, the government routed it through the vendors. R-6 (vendor) found the process 'unnecessarily complicated' as first, the vendors had to give the subsidy to the customers on installation and then avail the subsidy from the government. R-10 (vendor) pointed out that 'the whole subsidy thing was needlessly lengthy and can sometimes take even six months'. Since many of the vendors were small business owners with limited number of employees, they claimed that the 'delay' in availing the subsidy was hurting their cash flow. Moreover, some of them feared that after the subsidized installation, if the government officials find a fault in the installation, they could cancel the subsidy (R-9, vendor). Most of the vendors felt that the process can be streamlined by directly transferring the subsidy amount in the customers' account and by making the process easier by having as much of it online as possible.

5.5. Trust

Trust is commonly seen to be grounded in the perception of expertise, knowledge, honesty and openness of the messenger (Peters et al., 1997) and is a significant, yet underresearched, determinant of acceptance of technology and policies (Greenberg, 2014). It is an important heuristical element in environmental decision making process (Sovacool, 2014) and its importance is compounded in situations where there is uncertainty and lack of knowledge, like during introduction of a new technology in the society (Bronfman and Vázquez, 2011).

A lack of trust and a sense of suspicion was observed in the society members towards the vendors. Many resident respondents expressed doubts over vendors' intentions, especially with regards to sharing adequate information with them. Vendors were frequently accused of 'deliberately hiding facts' and 'not sharing the full information' (R-37, society member). In fact, R-37 categorically stated that the lack of trust was the main reason for 'not thinking about solar anymore'. Similarly, R-36 (society member) felt vendors were 'not sharing many details about solar with him'. He also was confused about the prices quoted by the vendors but his queries were unanswered. All this made him 'suspicious about the solar market'. R-26 (society member) expressed his wariness over the vendors' reasons for not giving him clear answers, while musing whether they were 'perhaps playing some game'. In a more accusatory tone, R-33 (society member) maintained that vendors 'were trying to fool people by not giving detailed documents'. In this situation he felt that it was impossible 'to trust them [vendors]'. R-21 (society member) stated that they were 'very much interested in solar' but couldn't move forward as they could not 'rely on vendors for giving full information as they [vendors] had their own interest'.

5.6. Other barriers

Other barriers were mostly related to the administrative issues of housing societies, chiefly, long time taken by them for decision making and risks perceived by the vendors in entering into a long term contract with CGHS. Several respondents (R-16, government/DISCOM official; R-20 government/DISCOM official and R-25, vendor) pointed out that the time housing societies take for decision making is too long, which induces uncertainty in the whole process. Also, most housing societies have elections every 1–2 years, which makes a long term contract signed in the RESCO model risky for the vendors (R-16, government/DISCOM official). There can be no iron clad guarantee that the newly elected members would always honour the previous members' decision (R-26, government/DISCOM official). Though the contract has the force of the law, R-25 (vendor) pointed out that taking legal recourse in case of any violation would itself be a painfully lengthy and expensive process. Moreover societies themselves are wary of signing a long term contract which according to them would hold their roof space 'hostage for 25 years' (R-16, government/DISCOM official). Another significant problem highlighted by R-26 (DISCOM/government official) was that as high paying commercial consumers switch to RTS, 'the utility will face two major issues'. First, in the DISCOM's tariff structure, almost 50% of the fixed charges are built into the tariff. On switch to RTS, fixed charges remain constant but the tariff payment to DISCOM may be reduced. However, since DISCOM's tariff has a significant amount of fixed charges built into it, DISCOM loses this fixed charge too, which ideally shouldn't have been effected by the switch to solar. Second, since domestic consumers are subsidized at the expense of high paying commercial customers, if latter switches to RTS, domestic consumers' billing burden may increase. He assured that 'the government is looking into these matters but one should not expect solutions overnight'.

6. Conclusion and policy implications

Delhi has both potential and need for the adoption of RTS. Delhi government's solar policy of 2016 was thus a timely step. However, the conceptualization and the implementation of the policy has not enabled the optimum diffusion of solar. A course correction is required at this stage for widespread adoption of RTS systems in Delhi. Few suggestions are mentioned below that would be helpful in this regard.

6.1. Launching awareness campaign and creating communication platforms

The relatively successful solar adoption in other states was due to a consistent and structured mass awareness campaign (Harish et al., 2013). Thus, a sustained mass awareness campaign clearly conveying the benefits of solar is needed. Cue can be taken from successful mass media campaigns in the past like the polio mass immunization program. Government should create platforms to facilitate frequent meetings between vendors and residents to remove the suspicion among society members and ensure a smooth flow of communication. On the vendors' front, communication channels must be established between them and government officials as vendors complained of the lack of platforms where they could meet officials face to face and voice their grievances.

6.2. Hassle free finance

Bank loan requirements like mortgage makes it difficult for entities like residential societies, who don't have any collective mortgage, to install solar. The solar financing process needs to be made more consumer friendly, more accessible, less cumbersome and if possible, specific loans tailored for housing societies' solar installations must be introduced. Another helpful intervention could be to encourage private finance companies to enter into the market to provide solar loans. These companies can collaborate with the vendors in formulating a business model that can be beneficial for all the stakeholders. Studies like Gutermuth (1998) have suggested some measures that can be taken by governments to make renewable energy more lucrative.

6.3. Streamlining the subsidy process

Vendors felt that there was too much time and documentation required to avail capital subsidy. Two main changes must be made in the subsidy availing process. Firstly, wherever possible, verifications and documentations should be made online to make the process hassle free. In the last few years the government has taken several steps to digitize India. It has issued Aadhaar cards, a unique identification number for each individual based on biometric details and linked them to bank accounts. Therefore, online disbursement of subsidies should not be difficult. Secondly, the subsidy should be made directly to the customers and not routed indirectly through the vendors.

6.4. Addressing bidding price grievances

'Unfairly' low bidding by few vendors by leveraging CSR funds was a source of serious discontent as other vendors felt that quoted price was even below breakeven. Regardless of the legality of the approach, government needs to engage with the disgruntled vendors and address the issue. Small solar businesses are an important part in promoting rooftop solar market in Delhi and if they would leave the market because of 'unviable' or 'unsustainable' bidding price, this would hurt solar market prospects. Some vendors suggested holding a periodic interactive platform comprising vendors and government officials where vendors could voice issues like above and come to mutually acceptable solutions. This can be a constructive step and the government should think about it.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Dwarkeshwar Dutt: Conceptualization, Data curation, Methodology, Writing - review & editing.

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